

Thesis Proposal: A Framework for Self-Adaptive Agentic Systems via Dynamic Capability Discovery and Composition in Enterprise Environments

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Proposed Start Date: Fall 2026

Date: Dec 14, 2025

1. Abstract

Modern agentic automation platforms promise to transform complex enterprise workflows by encapsulating domain expertise into reusable, specialized AI agents. However, their predominant modular architecture introduces a critical rigidity: the automation scope is bounded by pre-defined agent capabilities. Introducing support for novel tasks requires a manual, resource-intensive cycle of requirement analysis, tool development, and agent retraining, which fundamentally limits scalability and adaptability.

This research addresses the core challenge of **agentic adaptability**. We propose a novel framework for self-adaptive agentic systems, centered on two innovations: (1) a **Capability Knowledge Graph (CKG)** that provides a machine-readable, semantic map of all available tools, agents, APIs, and their functional profiles, and (2) a **Meta-Cognitive Orchestrator** that leverages this graph to dynamically decompose high-level user objectives, discover and sequence necessary capabilities, and integrate contextual knowledge from enterprise sources. The framework will be evaluated through a prototype simulating a complex enterprise automation environment (e.g., ERP system analysis and integration). Metrics will focus on the reduction in manual configuration effort, success rate on unseen composite tasks, and the system's ability to generate valid execution plans compared to static, module-centric baselines. This work aims to provide both a formal model for dynamic agent composition and an open-source architectural blueprint, advancing the field toward truly flexible and scalable enterprise AI.

2. Introduction & Background

The automation of enterprise systems—such as ERP configuration, data integration, and code analysis—remains a significant challenge due to complexity, domain specificity, and constant change. The emerging paradigm of "**agentic automation**" seeks to address this by deploying AI agents that act as intelligent co-pilots for specific tasks. These systems typically follow a modular, tool-based architecture where specialized agents (e.g., for document analysis, code scanning, workflow execution) operate on discrete sub-problems.

While effective for predefined tasks, this architecture suffers from a **static orchestration gap**. The intelligence is siloed within agents, but the higher-order logic of *when* and *how* to combine them for novel, composite objectives remains a manual human function. This limitation is evident in the industrial state-of-the-art, which requires extensive cross-functional effort to onboard new automation scenarios. The

core research problem, therefore, lies in transitioning from **closed-world, pre-programmed agent assemblies** to **open-world, dynamically composable agent systems**. This thesis investigates the mechanisms needed for agents to autonomously understand their collective capabilities and reorganize to meet emergent user goals, a key step toward realizing general-purpose assistant AI in the enterprise.

3. Problem Statement & Research Questions

Primary Problem: Current modular agentic systems lack a principled, autonomous mechanism for the dynamic discovery, composition, and execution of cross-functional capabilities in response to novel or complex user objectives. This limits their adaptability, scalability, and the realization of end-to-end automation.

Research Questions:

1. **RQ1 (Modeling & Discovery):** How can the diverse capabilities of an agentic ecosystem—including tools, APIs, and existing agents—be formally and semantically modeled to enable automated discovery and reasoning about their applicability?
2. **RQ2 (Planning & Composition):** Given a high-level natural language objective and a Capability Knowledge Graph, what algorithms can effectively generate and rank valid, executable workflows by dynamically sequencing discovered capabilities?
3. **RQ3 (Context Integration):** How can dynamic workflows be grounded and refined using contextual enterprise knowledge (e.g., data schemas, business glossaries, process diagrams) to improve their correctness and operational feasibility?
4. **RQ4 (Evaluation):** To what extent does a dynamic capability composition framework improve system adaptability and reduce the manual effort required to address novel tasks compared to a static agent architecture in enterprise automation benchmarks?

4. Literature Review

- **Agent-Oriented Software Engineering (AOSE) & Multi-Agent Systems (MAS):** Foundational work on agent communication, coordination, and organizational structures. This research will extend MAS principles to modern, LLM-powered agents with a focus on dynamic team formation for task-solving.
- **AI Planning & Automated Workflow Generation:** Classical planning (e.g., STRIPS, HTN) and modern neural-symbolic approaches. The thesis will contrast fixed, BPMN-style workflows with generative, context-aware planning over a graph of agent capabilities.
- **Knowledge Graphs for AI:** Research on using structured knowledge to enhance AI reasoning, including ontology design and graph-based retrieval. This work will specifically apply knowledge graphs to model agent capabilities and enterprise domain context.
- **Tool Learning & API Integration:** Studies on how LLMs can understand and use tools/APIs. This research will contribute a structured, graph-based approach to tool discovery and composition beyond single-tool invocation.
- **Industry Context:** While anonymized, the proposal is grounded in real-world challenges observed in state-of-the-art enterprise AI platforms, ensuring practical relevance.

5. Proposed Methodology

The research will follow a **design science** paradigm, creating and evaluating a novel artifact—the Self-Adaptive Agentic Framework.

Phase 1: Framework Design.

- **Capability Knowledge Graph (CKG) Schema:** Design an ontology to model agents, tools, and data sources. Each entity will be described by attributes such as action, input_schema, output_schema, preconditions, effects, and a natural language description.
- **Meta-Cognitive Orchestrator Architecture:** Design the core orchestrator with modules for (a) **Goal Decomposition** (using an LLM guided by the CKG), (b) **Capability Planning** (graph search/planning algorithms over the CKG), and (c) **Contextual Grounding** (querying enterprise knowledge graphs).
- **Simulation Environment:** Design a benchmark simulating a complex enterprise domain (e.g., ERP system management with modules for data, code, and process) populated with a set of discrete, tool-based agents.

Phase 2: Prototype Implementation.

- Implement the CKG using a graph database (e.g., Neo4j).
- Develop the Meta-Cognitive Orchestrator as a service, integrating an LLM for decomposition and a planner (e.g., based on hierarchical task network or graph traversal algorithms).
- Build the simulation environment with mock agents representing common enterprise functions (e.g., DocumentAnalyzer, SQLQueryEngine, CodeScanner, DataValidator).

Phase 3: Experimental Evaluation.

- **Benchmark:** Create a set of **composite enterprise tasks** of increasing complexity that require cross-agent coordination (e.g., "Analyze this error log, locate the responsible code module, check for known issues, and suggest a fix.").
- **Baseline:** A static, pre-wired workflow system representing current modular platforms.
- **Metrics:**
 - **Adaptability Success Rate:** Percentage of novel benchmark tasks successfully completed.
 - **Planning Efficiency:** Time and computational cost of dynamic workflow generation.
 - **Human Effort Reduction:** Measured by the decrease in manual steps required to configure the system for a new task.

6. Expected Contributions

- **Theoretical:** A formal model for dynamic capability discovery and composition in agentic systems, bridging AOSE, AI planning, and knowledge representation.
- **Architectural:** An open-source framework blueprint for building self-adaptive agentic systems, featuring the Capability Knowledge Graph and Meta-Cognitive Orchestrator.

- **Empirical:** A publicly available benchmark suite for evaluating adaptability in enterprise agentic AI, along with experimental results validating the proposed approach.

7. Project Plan & Timeline

- **Semester 1 (Fall 2026):** Finalize literature review. Detailed design of CKG ontology and orchestrator algorithms. Develop the benchmark task suite.
- **Semester 2 (Spring 2027):** Implement core framework components and the simulated agent environment.
- **Semester 3 (Summer 2027):** Conduct comprehensive experiments, analyze results, and begin thesis writing.
- **Semester 4 (Fall 2027):** Complete thesis writing, defend, and submit.

8. Preliminary References

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